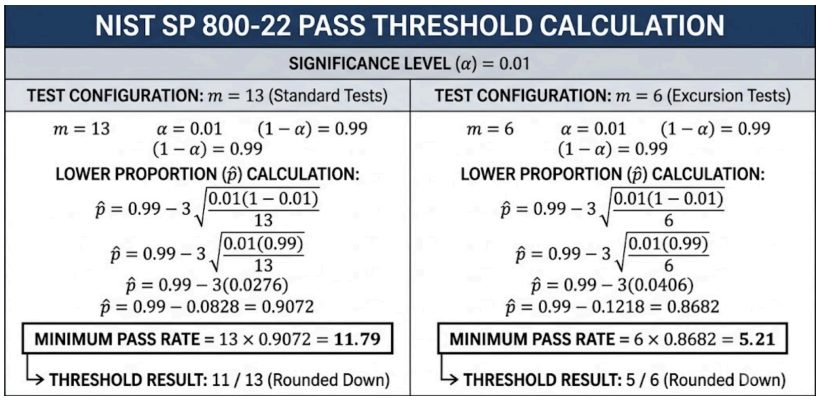
NIST Report

Input Format

1. ASCII Binary Format: There must be no spaces, no commas, and no newlines/line breaks between the characters. It must be a single, unbroken string of '0's and '1's.
2. Minimum Sample Size (Bit Length):
    a. Total Bits: To run the entire suite of 15 tests effectively, you generally need a massive sample size—ideally at least 10^6 to 10^7 bits (approx. 1 MB to 10 MB of raw binary data).
    b. Sequence Division (n): The suite will prompt you to divide your total bitstream into smaller, equal-sized blocks/sequences (n). A common standard for a single sequence length is n = 1,000,000 bits.
3. No Headers or Metadata

Test file format

1. File name: pure_ibm_quantum_master_13M.txt
2. Total Bits: The file contained approximately 13,000,000 bits (13 Megabits) of quantum random data.
3. How it was divided: The NIST software divided the file into 13 equal sequences. This means each sequence (or chunk) was exactly 1,000,000 bits (1 Megabit) long.
4. RandomExcursions: Considered only 6Mb of data
    a. Why it dropped: The Random Excursions test converts your binary data into a "random walk." By strict NIST mathematical rules, a data sequence must cross back to the starting point (zero) at least 500 times to even qualify for scoring.
    b. Not a failure: Because your numbers are truly random, 7 of the sequences naturally drifted far away from zero and simply ran out of bits before hitting 500 crossings. The software programmatically discards these because there isn't enough data to run the math—it does not mean they are non-random.
    c. The result: The remaining 6 sequences met the threshold and were tested . Passing them 6/6 mathematically proves your dataset behaves exactly like an ideal random generator .
5. Pass Threshold:
    a. For 13 sequences - 11 or more sequences are considered as pass
    b. For 6 sequences - 5 or more sequences are considered as pass



Verdict of each test cases

1. Frequency (Monobit) Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It counts the total number of 1s and 0s in your file to see if they are roughly equal (50/50). If a file has too many 1s or too many 0s, it fails.

2. Block Frequency Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It breaks your data into smaller blocks and checks if the balance of 1s and 0s is fair *inside* each block.

3. Cumulative Sums (Cusum) Test
    - Verdict: PASSED (Both sub-tests got 13/13).
    - What it checks: It travels through your data step-by-step, adding 1 for a '1' and subtracting 1 for a '0'. It checks if the running total wanders too far away from zero, which would reveal an accidental trend.

4. Runs Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It looks at how fast the numbers switch back and forth between 0 and 1. Switching too quickly or staying on one number too long triggers a fail.

5. Longest Run of Ones in a Block
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It checks the longest continuous streak of 1s inside smaller blocks to make sure the streaks aren't longer than what you would expect by pure random chance.

6. Binary Matrix Rank Test
    - Verdict: PASSED (12/13 sequences passed; minimum required was 11).
    - What it checks: It extracts small grids (matrices) from your bits and checks for repeating linear patterns or mathematical dependencies between rows.

7. Discrete Fourier Transform (FFT / Spectral) Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It looks for repeating patterns or periodic cycles across your entire data file (like a repeating rhythm).

8. Non-Overlapping Template Matching Test
    - Verdict: PASSED (All 148 template variations passed) .
    - What it checks: It searches your file for specific, short bit patterns (e.g., 000001). If certain patterns appear way too often or barely at all, it indicates non-random behavior.
    - There are a total of 148 templates of patterns to check by the NIST.

9. Overlapping Template Matching Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: Similar to the one above, but it looks for bursts of repeating short patterns that overlap each other (like 11111 searching for 111).

10. Maurer's "Universal Statistical" Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It checks how much the data can be compressed without losing information. Truly random data cannot be compressed significantly.

11. Approximate Entropy Test
    - Verdict: PASSED (13/13 sequences passed).
    - What it checks: It compares the frequency of short overlapping blocks of two adjacent lengths (like blocks of 2 bits vs blocks of 3 bits) to check for predictable patterns.

12. Random Excursions & Random Excursions Variant Tests
    - Verdict: PASSED (All sub-tests got 6/6; minimum required was 5) .
    - What it checks: These tests look at a "random walk" (adding 1s and subtracting 0s). They check if the running total returns to exactly zero 500 times, and how many times specific states are visited.

13. Serial Test
    - Verdict: PASSED (Both parts passed with 12/13 and 13/13).
    - What it checks: It checks if every possible combination of a specific length (like 00, 01, 10, 11) has an equal chance of appearing sequentially.
    - **First P-Value:** Checks if the distribution of all m-bit patterns is completely uniform (i.e., do all 2^m combinations show up roughly the same number of times?).
    - **Second P-Value:** Evaluates the consistency between the m-bit patterns, the (m-1)-bit patterns, and the (m-2)-bit patterns. This checks for transitional dependencies to ensure that a preceding bit doesn't secretly influence or predict the value of the next bit.

14. Linear Complexity Test
    - Verdict: PASSED (12/13 sequences passed).
    - What it checks: It determines how long a mathematical formula would need to be to perfectly mimic your sequence. If the data can be created with a short, simple formula, it isn't random.

Random Number Generating Process

1. random_number_generator.ipynb
2. System specification:
    a. IBM Kingston
    b. Heron r2 architecture
    c. Delivers up to 340,000 CLOPS (Circuit Layer Operations Per Second).
    d. Low Error Rates: Features an exceptionally low median two-qubit error rate of 2.03×10^{-3}.
    e. 156-qubit
3. Generates 1 million bits per file using 100 qubits. It applies a "Hadamard" gate to all of them, putting them into a perfect 50/50 quantum state to generate truly random numbers.
4. It runs in a loop getting 1 million bits per file. New file is created for the next 1 million bits. Example: sequence_0001.txt
5. As a plan B, some sequences were generated by local computer but were not used.
6. File stitcher: It takes the 13 files that successfully came from the real quantum computer, merges them together into one single giant file containing 13 million random bits, and saves it.